

THE UNIVERSITY OF LAHORE

Faculty of Information Technology | Department of Computer Science & IT
CS-13410 — Introduction to Machine Learning | Fall 2026 | BSCS

SEMESTER PROJECT — PART 1: PROJECT PROPOSAL

Predicting Stroke Risk from Patient Health Records

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1. Problem Statement

Stroke is one of the leading causes of death and long-term disability worldwide, and many patients are only identified after the event. Our project asks: given routine patient information (age, hypertension, heart disease, average glucose level, BMI, smoking status and basic background), can we predict whether a patient has had a stroke, and which factors matter most? A reliable risk-flagging model could help clinics direct screening and preventive care to the highest-risk patients first. The project is for learning only and is not meant for real clinical decisions.

2. Dataset

Name / source	Stroke Prediction Dataset by fedesoriano, published on Kaggle (kaggle.com/datasets/fedesoriano/stroke-prediction-dataset); released for educational use, credited to its author
Size	5,110 patient records, 12 columns (4,088 train / 1,022 test after a stratified 80/20 split)
Feature types	Continuous (age, avg_glucose_level, bmi), binary (hypertension, heart_disease, ever_married, Residence_type), nominal (gender, work_type, smoking_status), plus an anonymous id that is dropped
Data issues	201 missing BMI values (3.9%), 1,544 “Unknown” smoking entries, outliers in glucose and BMI, one rare gender category, severe class imbalance
Privacy / licensing	Patients are anonymous numbers; public Kaggle dataset for educational use with credit to the author

3. Target Variable and Task

Target	stroke: 1 if the patient had a stroke, 0 if not
Task type	Supervised binary classification
Class balance	Severely imbalanced: 4.9% stroke (249) vs 95.1% no stroke (4,861)
Success metrics	ROC-AUC and PR-AUC (main), plus recall and F1-score of the stroke class. Accuracy is not used because predicting “no stroke” for everyone already gives about 95%.

4. Why This Dataset Is Suitable

- **Meets the brief:** more than 1,000 rows with a mix of continuous, binary and categorical features.
- **Real preprocessing work:** informative missing values, hidden “Unknown” categories, outliers and severe class imbalance.
- **Room to grow across all five parts:** tree and probabilistic models (Part 2), ensembles and tuning (Part 3), a neural network, clustering/PCA of patient profiles and a fairness discussion (Part 4).
- **Easy to work with:** one small, well-documented CSV with a clear medical story every group member can explain in the viva.

5. Part 1 Plan: Leakage-Free Pipeline

- **Audit and split:** check missing values, duplicates and ranges, then make a stratified 80/20 train/test split before fitting anything.
- **Preprocess with Pipeline and ColumnTransformer:** median/most-frequent imputation, 3×IQR outlier capping, standard scaling and one-hot encoding, all fitted on training data only.
- **Features:** construct bmi_missing, is_senior and comorbidity_count; select the top 18 features with the ANOVA F-test.
- **Imbalance and EDA:** stratified CV with class weights; seven visualisation sections in the notebook.